Bounded Dyadic Profiles with Full Absolute Achievement Set

FA/Banach automatic math run: `fa_banach_001`

2026-07-19

Status

Partial result likely valid. Marchwicki and Vlasák ask for a characterization of the functions f for which

$$\mathcal{A}_{\text{abs}}\left(\frac{(-1)^n}{n}, \frac{(-1)^n}{f(n)}\right) = \mathbb{R}^2.$$

They prove this for $f(n) = n^\alpha$, $0 < \alpha < 1$ [1]. The theorem below gives a large non-power family: the coefficient in the second coordinate may be changed by an arbitrary bounded positive function of the odd part of n. An explicit two-valued profile is then shown to give a genuinely conditionally convergent vector series, not merely a sequence with full absolute achievement set.

Source Question

The two characterization problems appear on page 6 of arXiv:1802.10535 [1]:

$$\Psi(W, (\alpha_1, \dots, \alpha_d)) = \sum_{k=1}^{d} \Psi(W^k, (\alpha_1, \dots, \alpha_d)) = (x^1, \dots, x^d)$$

and we are done.

□

3. Open problems

Problem 3.1. *Characterize a familly of all functions f such that* $\mathrm{A}_{\text{abs}}((\frac{(-1)^n}{n}, \frac{(-1)^n}{f(n)}) = \mathbb{R}^2$.

Problem 3.2. *Characterize a familly of all functions f such that* $\mathrm{A}((\frac{(-1)^n}{n}, \frac{(-1)^n}{f(n)}) = \mathbb{R}^2$.

Clearly the familly defined in the first problem is smaller than that defined in the second problem. In the paper we showed that it contains any function $f(n) = n^\alpha$ for $\alpha \in (0, 1)$.

Transcribed, the printed questions are:

Problem 3.1. Characterize a family of all functions f such that

$$\mathcal{A}_{\text{abs}}\left(\frac{(-1)^n}{n}, \frac{(-1)^n}{f(n)}\right) = \mathbb{R}^2.$$

Problem 3.2. Characterize a family of all functions f such that

$$\mathcal{A}\left(\frac{(-1)^n}{n}, \frac{(-1)^n}{f(n)}\right) = \mathbb{R}^2.$$

The present packet partially answers Problem 3.1. Since $\mathcal{A}_{\text{abs}} \subset \mathcal{A}$, it also supplies examples for Problem 3.2.

Definitions and Statement

For $n \geq 1$, let

$$\mathrm{odd}(n) = \frac{n}{2^{v_2(n)}}$$

be the odd part of n. For a vector sequence (u_n) in a finite-dimensional normed space, write

$$\mathcal{A}_{\mathrm{abs}}(u_n) = \left\{ \sum_{n \in S} u_n : S \subset \mathbb{N},\ \sum_{n \in S} \|u_n\| < \infty \right\}.$$

The particular norm is immaterial.

Proof Intuition

The key structure is that $a(\mathrm{odd}(n))$ is constant along each dyadic ray $\{2^j k : j \geq 0\}$ with k odd. Choose dyadic levels whose $2^{-\alpha j}$-weights sum to one. Adding those negative even terms to the positive odd term cancels the second coordinate exactly but leaves a positive horizontal displacement. A sparse exact harmonic-subsums lemma selects odd rays giving any prescribed horizontal displacement while keeping the total α-cost finite. Conversely, summing an entire dyadic ray cancels the first coordinate and leaves a negative vertical displacement; a greedy subsum construction fills any prescribed negative height. These two constructions reach the southeast quadrant beyond every tail, and a finite subsum moves an arbitrary target into that quadrant. This yields a broad sufficient family, but the full characterization requested in Problems 3.1–3.2 remains open.

Theorem 1 (bounded odd-part profiles)**.** *Fix $0 < \alpha < 1$. Let a be any function on the positive odd integers such that*

$$0 < m \leq a(k) \leq M < \infty.$$

Define

$$u_n = (-1)^{n+1} \left(\frac{1}{n}, \frac{a(\mathrm{odd}(n))}{n^\alpha} \right).$$

Then

$$\mathcal{A}_{\mathrm{abs}}(u_n) = \mathbb{R}^2.$$

Equivalently, Problem 3.1 has a positive answer for every

$$f(n) = \frac{n^\alpha}{a(\mathrm{odd}(n))}$$

with a bounded above and bounded away from zero.

The sign in the source is $(-1)^n$ rather than $(-1)^{n+1}$. Multiplying the whole sequence by -1 does not change the conclusion.

Two Elementary Subsum Lemmas

Lemma 1 (sparse exact harmonic sums)**.** *Let $0 < \alpha < 1$. Given $t > 0$ and $N_0 \in \mathbb{N}$, there is a set H of odd integers, all greater than N_0, such that*

$$\sum_{k \in H} \frac{1}{k} = t, \qquad \sum_{k \in H} \frac{1}{k^\alpha} < \infty.$$

Proof. Choose numbers $1 < \lambda < \rho < 2$. Starting sufficiently far out, choose an increasing sequence of odd integers $q_j > N_0$ with

$$\lambda q_j \le q_{j+1} \le \rho q_j.$$

For $Q = \{q_j : j \ge 1\}$, both $\sum_Q k^{-1}$ and $\sum_Q k^{-\alpha}$ converge by the lower geometric bound. Also

$$\sum_{\ell > j} \frac{1}{q_\ell} \ge \frac{1}{q_j} \sum_{s \ge 1} \rho^{-s} = \frac{1}{(\rho - 1) q_j} > \frac{1}{q_j}.$$

Hence the standard greedy subsum argument gives

$$\left\{ \sum_{k \in E} \frac{1}{k} : E \subset Q \right\} = [0, \delta], \qquad \delta := \sum_{k \in Q} \frac{1}{k}.$$

Indeed, for a decreasing positive summable sequence whose every term is at most its remaining tail, greedily including a term whenever it does not overshoot a prescribed target leaves zero remainder.

The harmonic series over the odd integers outside Q, and beyond any fixed threshold, still diverges. If $t > \delta$, take a finite set G in that complement such that

$$t - \delta \le \sum_{k \in G} \frac{1}{k} \le t;$$

this is obtained by starting where every term is smaller than δ and stopping at the first crossing. If $t \le \delta$, take $G = \varnothing$. Choose $E \subset Q$ whose reciprocal sum is $t - \sum_G k^{-1}$, and put $H = E \cup G$. The desired harmonic identity holds, while the α-sum is finite because G is finite and $E \subset Q$. $\square$

Lemma 2 (exact subsums of a divergent null sequence)**.** *If $b_j > 0$, $b_j \to 0$, and $\sum_j b_j = \infty$, then for every $y \ge 0$ there is $J \subset \mathbb{N}$ such that $\sum_{j \in J} b_j = y$.*

Proof. Starting with remainder y, include b_j exactly when it does not exceed the current remainder. If the limiting remainder were positive, then eventually every b_j would be smaller than it; divergence would force the remainder below zero, a contradiction. $\square$

Proof of the Profile Theorem

Proof of Theorem 1. Put $r = 2^{-\alpha}$. Since $1/2 < r < 1$, the same no-gap subsum argument as in Lemma 1 gives a set $J \subset \mathbb{N}$ satisfying

$$\sum_{j \in J} r^j = 1.$$

Define

$$c = 1 - \sum_{j \in J} 2^{-j}.$$

Then $c > 0$: the last sum is at most 1, and equality would force $J = \mathbb{N}$, whereas $\sum_{j \ge 1} r^j = r/(1-r) > 1$.

For an odd integer k, consider the dyadic block

$$B_k = \{k\} \cup \{2^j k : j \in J\}.$$

The odd term has positive sign and all the other terms have negative sign. Because $\operatorname{odd}(2^j k) = k$, exact dyadic self-similarity gives

$$\sum_{n \in B_k} u_n = \left(\frac{1}{k} \left(1 - \sum_{j \in J} 2^{-j} \right), \frac{a(k)}{k^\alpha} \left(1 - \sum_{j \in J} 2^{-\alpha j} \right) \right) = \left(\frac{c}{k}, 0 \right). \tag{1}$$

Moreover, using the ℓ^1-norm on $\mathbb{R}^2$,

$$\sum_{n \in B_k} \|u_n\|_1 \le \frac{2}{k} + \frac{2M}{k^\alpha}. \tag{2}$$

Now fix $x > 0$ and a tail threshold N_0. Lemma 1 provides odd $H \subset (N_0, \infty)$ such that

$$\sum_{k \in H} \frac{1}{k} = \frac{x}{c}, \qquad \sum_{k \in H} k^{-\alpha} < \infty.$$

Distinct odd k's generate disjoint dyadic blocks. Thus (1)–(2) show that the union of the B_k, $k \in H$, is an absolutely summable subseries with sum $(x, 0)$.

For vertical motion use the entire dyadic ray

$$C_k = \{2^j k : j \ge 0\}, \qquad k \text{ odd}.$$

Its sum is

$$\sum_{n \in C_k} u_n = \left(0, \frac{a(k)}{k^\alpha} \left(1 - \sum_{j \ge 1} r^j \right) \right) = \left(0, -\kappa \frac{a(k)}{k^\alpha} \right), \tag{3}$$

where

$$\kappa = \frac{2r - 1}{1 - r} > 0.$$

The total absolute cost of C_k is bounded by a constant depending only on α, m, M times $k^{-\alpha}$.

Remove the preceding set H from the odd integers beyond N_0. On the remaining set the positive terms $a(k)k^{-\alpha}$ tend to zero and have divergent sum: the full odd α-series diverges, while $\sum_H a(k)k^{-\alpha} < \infty$. Given $y < 0$, Lemma 2 therefore supplies a set U, disjoint from H, such that

$$\sum_{k \in U} \frac{a(k)}{k^\alpha} = \frac{-y}{\kappa}.$$

Equation (3) gives sum $(0, y)$. It is an absolutely summable subseries, since

$$\sum_{k \in U} k^{-\alpha} \le \frac{-y}{m\kappa} < \infty \quad \text{and} \quad k^{-1} \le k^{-\alpha}.$$

Again the dyadic rays are mutually disjoint. Consequently, every point in the quadrant

$$\{(x, y) : x > 0, y < 0\} \tag{4}$$

is realized by an absolutely convergent subseries using indices beyond any prescribed threshold.

It remains to move an arbitrary target $z = (z_1, z_2)$ into (4) by a finite subsum. First choose finitely many even indices E so that their first coordinate sum s_1 is less than $z_1 - 1$; this is possible because the negative even harmonic subseries diverges. Let s_2 be their second coordinate sum, and choose

$$D > \max\{1, z_2 - s_2\}.$$

Far enough out on the odd indices, the ratio of the second positive coordinate to the first is

$$a(n)n^{1-\alpha} \geq mn^{1-\alpha},$$

which is arbitrarily large. Choose a tail on which this ratio is greater than $D+1$ and every second-coordinate term is less than 1. Since the positive odd second-coordinate subseries diverges, a finite set O in that tail has second-coordinate sum in $[D, D+1]$. Its first-coordinate sum is then less than 1. Hence for the finite set $F = E \cup O$,

$$\left(z - \sum_{n\in F} u_n\right)_1 > 0, \qquad \left(z - \sum_{n\in F} u_n\right)_2 < 0.$$

Apply the tail version of (4) beyond $\max F$ to realize this residual. Together with F, this gives an absolutely convergent subseries summing to z. Since $z \in \mathbb{R}^2$ was arbitrary, the theorem follows. $\square$

A Genuinely Conditionally Convergent Non-Power Family

The profile theorem by itself does not require the full vector series to converge. The next corollary gives an explicit non-power family in the conditionally convergent setting of the source paper.

Corollary 1. *Fix $0 < \alpha < 1$ and constants $A, B > 0$. For odd k, put*

$$a(k) = \begin{cases} A, & k \equiv 1 \pmod 4, \\ B, & k \equiv 3 \pmod 4. \end{cases}$$

Then the vector series

$$\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{1}{n}, \frac{a(\mathrm{odd}(n))}{n^\alpha}\right)$$

converges conditionally and its absolute achievement set is $\mathbb{R}^2$. If $A \neq B$, the corresponding function $f(n) = n^\alpha / a(\mathrm{odd}(n))$ is not a constant multiple of a power.

Proof. Only convergence of the second coordinate needs proof. Put $d_n = a(\mathrm{odd}(n))$, $T(N) = \sum_{n\leq N} d_n$, and let $P(N)$ be the same sum restricted to odd n. The values on consecutive odd integers alternate between A and B, so

$$P(N) = \frac{A+B}{4}N + O(1).$$

Since $d_{2n} = d_n$,

$$T(N) = T(\lfloor N/2 \rfloor) + P(N) = T(\lfloor N/2 \rfloor) + \frac{A+B}{4}N + O(1).$$

Iterating this identity and summing the dyadic floors gives

$$T(N) = \frac{A+B}{2}N + O(\log N).$$

Therefore the partial sums of the unweighted signed profile satisfy

$$\sum_{n\leq N}(-1)^{n+1}d_n = T(N) - 2T(\lfloor N/2 \rfloor) = O(\log N).$$

Summation by parts now shows that $\sum(-1)^{n+1}d_n n^{-\alpha}$ converges, because $n^{-\alpha} - (n+1)^{-\alpha} = O(n^{-\alpha-1})$ and $\sum(\log n)n^{-\alpha-1} < \infty$. It is not absolutely convergent, since $d_n \geq \min(A, B) > 0$ and $\sum n^{-\alpha} = \infty$. The first coordinate is the alternating harmonic series. Thus the vector series is conditionally convergent, and Theorem 1 gives its absolute achievement set. $\square$

For example, one may take $A = 1$ and $B = 2$. This gives the concrete non-power function

$$f(n) = \begin{cases} n^{\alpha}, & \mathrm{odd}(n) \equiv 1 \pmod 4, \\ \frac{1}{2}n^{\alpha}, & \mathrm{odd}(n) \equiv 3 \pmod 4. \end{cases}$$

Scope, Novelty Check, and Review Notes

This does not characterize all functions in Problems 3.1–3.2. It isolates one robust sufficient mechanism: the second coordinate may vary arbitrarily between dyadic rays, provided it has exact power scaling along each ray and the ray profiles stay uniformly comparable. The proof is independent of the prime-factor induction used for the source paper's higher-dimensional power theorem.

The local run indexes were searched for arXiv:1802.10535, its title, achievement sets, dyadic profiles, odd parts, and the displayed function family. No duplicate was found. Bounded web searches on 2026-07-19 checked the source paper and its two immediate predecessor papers [2, 3]; no bounded odd-part-profile statement was found.

Human review should focus on three elementary points: the exact sparse harmonic subsum lemma, disjointness of dyadic rays indexed by distinct odd parts, and the final finite-subsum shift into the southeast quadrant. The conditional-convergence corollary is logically separate from the achievement set construction.

References

[1] J. Marchwicki and V. Vlasák. *Subsums of conditionally convergent series in finite dimensional spaces*. arXiv:1802.10535, 2018.

[2] A. Bartoszewicz, S. Głab, and J. Marchwicki. *Achievement sets of conditionally convergent series*. Colloquium Mathematicum 152 (2018), 235–254; arXiv:1604.07575.

[3] S. Głab and J. Marchwicki. *Lévy–Steinitz theorem and achievement sets of conditionally convergent series on the real plane*. Journal of Mathematical Analysis and Applications 459 (2018), 476–489; arXiv:1705.06472.